Tuesday, August 23, 2022
Duration: 180 minutes
Total: 50 pts
Instructor: Dr. Ali Farid

Student Name:………………………………..

Student Number: …………………………….

Signature: …………………………………….

Exercise 1:	**/4**	**pts**
Exercise 2:	**/8**	**pts**
Exercise 3:	**/11**	**pts**
Exercise 4:	**/12**	**pts**
Exercise 5:	**/4**	**pts**
Exercise 6:	**/4.5**	**pts**
Exercise 7:	**/6.5**	**pts**
Total:	**/50**	**pts**

Notes:

1. This is a closed book exam with no electronic devices allowed
2. Write your Name and Student Number, and sign this first page
3. Write your name on each of the 10 pages plus the cover page
4. Answer all questions in the space provided or use the back side of the sheet as needed
5. You must submit this examination booklet and sign the attendance sheet
6. You must maintain the confidentiality of your examination; do not provide any opportunity for others to copy any of your work

Exercise 1: Short questions [1+1+1+1=4pts]

Define the following concepts in your own words.

- Combinatorial problem

- Feasible solution versus optimal solution.

- Dynamic Programming versus Divide and Conquer algorithm

- Basic External Sorting Algorithm

Exercise 2: Algorithm Design Techniques [0.5x4+0.5x6+3=8pts]

When introducing the different algorithm design techniques, we focused on the following five common types of algorithms:

• Greedy algorithm

• Divide and Conquer

• Dynamic Programming

• Randomized Algorithms

• Backtrack Search

1. To which of the above types does each of the following algorithms belong to?

(a) Huffman Coding

(b) Prim's Algorithm

(c) Branch and Bound Algorithm

(d) Dijkstra's Shortest Path Algorithm

2. Which design technique can be used to solve each of the following problems?

(a) The loading problem

(b) Closest-Points Problem in O (N log N)

(c) Primality Testing using Fermat's Lesser Theorem

(d) The 0/1 Knapsack problem

(e) N Queens Problem

(f) Ordering Matrix Multiplications

3. Consider the following investment optimization problem:

The goal is to find the best investment (combination of investment options maximizing the total profit) for $15,000, given the following investment options.

Option 1: This investment requires $3,000 and gives a profit of $200.

Option 2: This investment requires $4,000 and gives a profit of $300.

Option 3: This investment requires $10,000 and gives a profit of $1,200.

Option 4: This investment requires $12,000 and gives a profit of $1,400

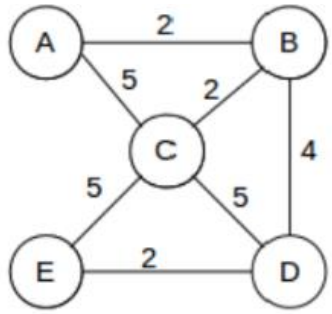
(a) What type of problem that was discussed in this course is similar to the above problem?

(b) What types of algorithms are the best when we want to have a fast algorithm?

(c) What types of algorithms are the best when we want to have the best solution regardless of computational time?

Exercise 3: Branch-and-Bound Algorithm [0.5+2+2+3.5+1+1+1=11 pts]

We want to apply the branch-and-bound algorithm to solve the traveling salesman problem for the following graph.



a) Describe the traveling salesman problem.

b) What is the lower bound to the above TSP graph?

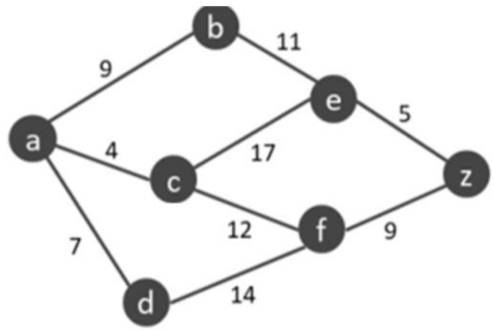
c) What is a solution tree for the above TSP graph?

d) What is a lower bound for each case in a solution tree for the above TSP graph?

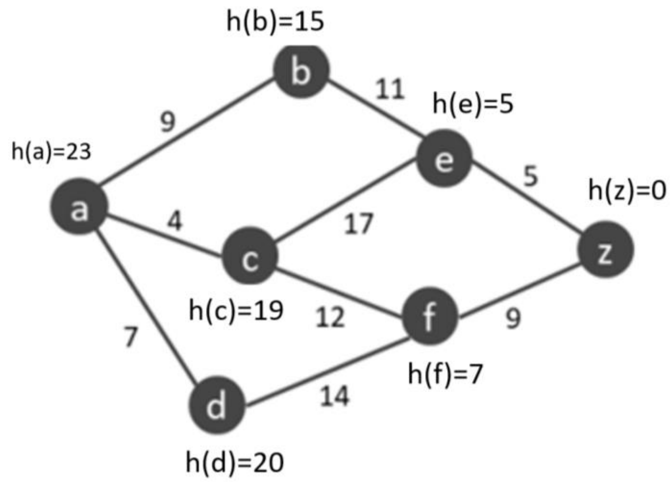
e) What is the optimal solution (shortest round trip) to this TSP problem?

f) If we remove edge (B,D) then is there any solution why?

Exercise 4: Dijkstra and A* Algorithms [5+1+6=12pts]

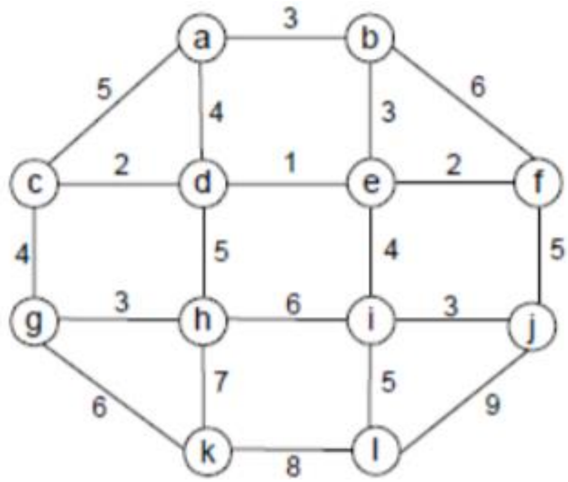


1. Implement Dijkstra's Algorithm and evaluate it for the graph above, and display the shortest distance from a to z?
2. What is the running time of Dijkstra's Algorithm?



1. Given the weights and heuristic values for the graph above, what would A* search return as the shortest path from a to z?

Exercise 5: Spanning Tree [4pts]

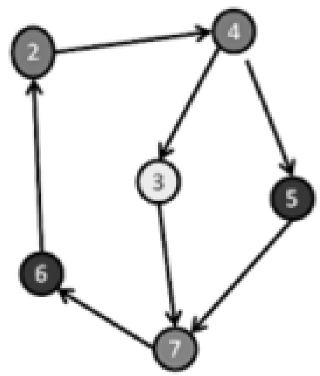


1. What is the Minimum Spanning Tree of the following graph for the above graph using Kruskal?

2. What is the Maximum Spanning Tree?

Exercise 6: Graphs [4.5 pts]

1. What is the topological order of this graph?



2. Answer the following questions as either True or False and provide a brief explanation:

a) If a graph with n vertices has n−1 edges, it must be a tree.

b) The adjacency matrix representation is typically better than the adjacency list representation when the graph is very connected.

c) Every edge is looked at exactly twice in every iteration of DFS on a connected, undirected graph.

d) Given a fully connected, directed graph (a directed edge exists between every pair of vertices), a topological sort can never exist.

Exercise 7: Sorting [2+0.5+2*0.5+1+2=6.5 pts]

1. Give an algorithm to sort 4 elements in 5 comparisons. Prove that this algorithm is optimal.

2. Given 64 unsorted data, and you are using Merge Sort to sort them. How many logical operations will be needed to sort them?
   a) About 60 b) About 400 c) About 2000 d) About 4000

3. True/False Questions:

   (a) Radix Sort can be used to sort integers and strings.

   (b) Heap Sort is always slower than quick sort.

4. Sort these algorithms based on their worst-case complexity:

   Radix sort, Merge Sort, Quick Sort, Bubble Sort, Insertion Sort

5. Suppose that the splits at every level of the quicksort algorithm are in the proportion 10/100 to 90/100. What is the maximum depth of a leaf in the recursion tree?